Gerui Wang

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Education

May 2022 – Aug. 2017	Ph.D. in Computer Science | Advisor: Prof. Pramod Viswanath Department of Computer Science, **University of Illinois Urbana-Champaign**
Jul. 2017 – Aug. 2013	Bachelor of Engineering in Computer Science & Technology Institute for Interdisciplinary Information Sciences, Yao Class **Tsinghua University**

Experience

Now – May 2022	Blockchain Research Scientist *Beijing Academy of Blockchain and Edge Computing, Beijing, China* Keywords: Blockchain, Research & Development
May 2022 – May 2021	Algorithm Engineer & Researcher *Hash Laboratories Pty. Ltd., North Sydney, NSW, Australia* Keywords: Blockchain, Research & Development
May 2022 – Aug. 2017	Research Assistant, advisor: Prof. Pramod Viswanath *University of Illinois Urbana-Champaign* Keywords: Blockchain, Consensus, Distributed System
Spring 2020	Teaching Assistant *University of Illinois Urbana-Champaign* Course: Principles of Blockchains
Summer 2019	Research Intern, advisor: Prof. David Tse *Applied Protocol Research, Palo Alto, CA, USA* Keywords: Blockchain, Consensus, Distributed System
Spring 2016	Visiting Student *Department of Computer Science, Cornell University*

Honors and Awards

- Beijing Model Worker
- ACM CCS'21 Best Paper Award Runner-up

Publications

[1] C. Hu, T. Xiao, J. Gao, Y. Zhao, Y. Li, X. Zhu, S. Liu, **G. Wang**, M. Wan, J. Wang, and J. Liu. "Detrax: Deterministic Parallel Transaction Scheduling for the ChainMaker Blockchain". In: *Blockchain: Research and Applications* (2026).

[2] Y. Hu, J. Liu, B. Zhao, Z. Pan, A. Liu, H. Tan, **G. Wang**, M. Wan, and Y. Liu. "A Heterogeneous Sharding Architecture for Privacy-Preserving Consortium Blockchains". In: *Proceedings of the International Conference on Knowledge Science, Engineering and Management (KSEM)*. 2026.

[3] Y. Liu, M. Shen, Y. Su, Y. Xie, W. Wang, Q. Zhao, J. Wang, J. Dong, **G. Wang**, and L. Zhu. "Enabling Verifiable Keyword Search Over Encrypted Data via Dynamic Version-Aware Indexing". In: *IEEE Transactions on Network Science and Engineering* 13 (2026).

[4] Z. Pan, J. Liu, A. Liu, Y. Hu, B. Zhao, Z. Guan, **G. Wang**, M. Wan, and Y. Liu. "Poros: Secure and Highly Parallel Sharding Blockchain with Optimized Cross-Shard Transaction Processing". In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*. CCS'26. 2026.

[5] P. Sheng, **G. Wang**, K. Nayak, S. Kannan, and P. Viswanath. "Player-replaceability and forensic support are two sides of the same (crypto) coin". In: *International Conference on Financial Cryptography and Data Security*. Springer. 2023, pp. 56–74.

[6] M. Fitzi, X. Wang, S. Kannan, A. Kiayias, N. Leonardos, P. Viswanath, and **G. Wang**. "Minotaur: Multi-Resource Blockchain Consensus". In: *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. CCS'22. 2022.

[7] P. Sheng*, **G. Wang***, K. Nayak, S. Kannan, and P. Viswanath. "BFT Protocol Forensics". In: *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*. CCS'21. 2021.

[8] **G. Wang**, S. Wang, V. Bagaria, D. Tse, and P. Viswanath. "Prism Removes Consensus Bottleneck for Smart Contracts". In: *2020 Crypto Valley Conference on Blockchain Technology (CVCBT)*. 2020, pp. 68–77. DOI: 10.1109/CVCBT50464.2020.00011.

[9] G. Fanti, L. Kogan, S. Oh, K. Ruan, P. Viswanath, and **G. Wang**. "Compounding of Wealth in Proof-of-Stake Cryptocurrencies". In: *Financial Cryptography and Data Security*. 2019, pp. 42–61. ISBN: 978-3-030-32101-7.

[10] Q. Zhao, **G. Wang**, X. Yuan, and X. Ma. "Efficient and robust detection of multipartite Greenberger-Horne-Zeilinger-like states". In: *Phys. Rev. A* 99 (5 May 2019), p. 052349. DOI: 10.1103/PhysRevA.99.052349.

[11] C. Li, **G. Wang**, and G. De Melo. "Context-Based Few-Shot Word Representation Learning". In: *2018 IEEE 12th International Conference on Semantic Computing (ICSC)*. IEEE. 2018, pp. 239–242.

* indicates equal contributions

Projects

- ChainMaker
  A leading open-source consortium blockchain platform
  `https://git.chainmaker.org.cn/chainmaker`
- XDC Consensus Engine DPoS 2.0
  Upgrading XDC network consensus to XDPoS 2.0
  `https://arxiv.org/abs/2108.01420`
- Prism Smart Contracts
  Design and implementation of a smart contract full-node client with the Prism consensus protocol
  `https://github.com/wgr523/prism-smart-contracts`
- Prism: Scaling Bitcoin by 10,000x
  Design and implementation of a novel scalable blockchain system in Rust language
  `https://github.com/yangl1996/prism-rust`
- Principles of Blockchains (a teaching assistant project)
  A project that helps students to build blockchain clients from scratch
  `https://gitlab.engr.illinois.edu/ece598pv/ece598pv-sp2022`

Computer Skills

Rust, Go, Python

Languages

English: Fluent
Chinese: Mother tongue